Práctica 2.1: Despliegue de NGINX + PHP-FPM

Actualizar Ubuntu

```
david@Despliegue:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://es.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://es.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://es.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:5 https://download.docker.com/linux/ubuntu resolute InRelease
5 packages can be upgraded. Run 'apt list --upgradable' to see them.
david@Despliegue:~$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-34          linux-main-modules-zfs-7.0.0-34-generic  linux-tools-7.0.0-34-generic
  linux-headers-7.0.0-34-generic  linux-modules-7.0.0-34-generic
  linux-image-7.0.0-34-generic    linux-tools-7.0.0-34
Use 'sudo apt autoremove' to remove them.

Not upgrading yet due to phasing:
  vim  vim-common  vim-runtime  vim-tiny  xxd

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 5
david@Despliegue:~$
```

1. Instalar PHP y extensiones (sudo apt install -y php8.5 php8.5-fpm php-cli php-mbstring php-xml php-curl)

```
david@Despliegue:~$ sudo apt install -y php8.5 php8.5-fpm php-cli php-mbstring php-xml php-curl
[sudo: authenticate] Password:
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-34          linux-main-modules-zfs-7.0.0-34-generic  linux-tools-7.0.0-34-generic
  linux-headers-7.0.0-34-generic  linux-modules-7.0.0-34-generic
  linux-image-7.0.0-34-generic    linux-tools-7.0.0-34
Use 'sudo apt autoremove' to remove them.

Installing:
  php-cli  php-curl  php-mbstring  php-xml  php8.5  php8.5-fpm

Installing dependencies:
  libargon2-1  php-common  php8.5-cli  php8.5-common  php8.5-curl  php8.5-mbstring  php8.5-readline  php8.5-xml

Suggested packages:
  php-pear

Summary:
  Upgrading: 0, Installing: 14, Removing: 0, Not Upgrading: 5
  Download size: 7,989 kB
```

Comprobar el estado de PHP-FPM

```
david@Despliegue:~$ sudo systemctl status php8.5-fpm
● php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager
     Loaded: loaded (/usr/lib/systemd/system/php8.5-fpm.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-11-05 17:41:53 UTC; 1min 25s ago
 Invocation: af6509b855574311a39fbf57ed89947c
       Docs: man:php-fpm8.5(8)
    Process: 34003 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.5/fpm/pool>
   Main PID: 33999 (php-fpm8.5)
     Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0.00req/sec"
      Tasks: 3 (limit: 4889)
     Memory: 10.4M (peak: 11.8M)
        CPU: 42ms
     CGroup: /system.slice/php8.5-fpm.service
             ├─33999 "php-fpm: master process (/etc/php/8.5/fpm/php-fpm.conf)"
             ├─34001 "php-fpm: pool www"
             └─34002 "php-fpm: pool www"

Nov 05 17:41:53 Despliegue systemd[1]: Starting php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager...
Nov 05 17:41:53 Despliegue systemd[1]: Started php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager.
lines 1-18/18 (END)
```

Comprobar las versiones de php y nginx

```
david@Despliegue:~$ php -v
PHP 8.5.4 (cli) (built: Oct 19 2026 14:35:18) (NTS)
Copyright (c) The PHP Group
Built by Ubuntu
Zend Engine v4.5.4, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.4, Copyright (c), by Zend Technologies
david@Despliegue:~$ nginx -v
nginx version: nginx/1.28.3 (Ubuntu)
david@Despliegue:~$
```

3. Conectar al servidor con SFTP

```
C:\Users\2DAW>sftp david@192.168.88.38
david@192.168.88.38's password:
subsystem request failed on channel 0
Connection closed

C:\Users\2DAW>
```

Instalar el servidor SSH en Ubuntu

```
david@Despliegue:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.2p1-2ubuntu3.8).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-34          linux-main-modules-zfs-7.0.0-34-generic  linux-tools-7.0.0-34-generic
  linux-headers-7.0.0-34-generic  linux-modules-7.0.0-34-generic
  linux-image-7.0.0-34-generic    linux-tools-7.0.0-34
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 5
```

Indicar a Ubuntu dónde se encuentra el programa que permite utilizar SFTP para transferir archivos

```
david@Despliegue: ~
  GNU nano 8.7.1                                  /etc/ssh/sshd_config
Subsystem sftp /usr/lib/openssh/sftp-server
```

Ya funciona correctamente

```
C:\Users\2DAW>sftp david@192.168.88.38
david@192.168.88.38's password:
Connected to 192.168.88.38.
sftp>
```

```
PS C:\Users\2DAW> sftp david@192.168.88.38
david@192.168.88.38's password:
Connected to 192.168.88.38.
sftp> mkdir /tmp/running-zones
remote mkdir "/tmp/running-zones": Failure
sftp> cd/tmp/running-zones
Invalid command.
sftp> cd /tmp/running-zones
```

```
sftp> lcd "C:/Users/2DAW/Desktop/tareas/Despliegue/Tema 2/running-zones"
sftp> put -r 04ArrayBootstrap.php
Uploading 04ArrayBootstrap.php to /tmp/running-zones/04ArrayBootstrap.php
04ArrayBootstrap.php
sftp> ls
```

4. Instalación de rsync

```
david@Despliegue:~$ sudo apt install rsync -y
[sudo: authenticate] Password:
rsync is already the newest version (3.4.1+ds1-7ubuntu0.3).
rsync set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-34          linux-main-modules-zfs-7.0.0-34-generic  linux-tools-7.0.0-34-generic
  linux-headers-7.0.0-34-generic  linux-modules-7.0.0-34-generic
  linux-image-7.0.0-34-generic    linux-tools-7.0.0-34
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 5
david@Despliegue:~$
```

crear el directorio web

```
david@Despliegue:~$ sudo mkdir -p /var/www/running-zones
david@Despliegue:~$
```

copiar los archivos

```
david@Despliegue:~$ sudo rsync -av /tmp/running-zones/* /var/www/running-zones/
sending incremental file list
Instalación y configuración del servidor web NGINX.odt
Instalación y configuración del servidor web NGINX.pdf
Práctica 2.1 Despliegue de NGINX  PHP FPM.odt

sent 968,243 bytes  received 73 bytes  1,936,632.00 bytes/sec
total size is 967,708  speedup is 1.00
david@Despliegue:~$
```

Configurar los permisos de la aplicación

```
david@Despliegue:~$ sudo chown -R www-data:www-data /var/www/running-zones
[sudo: authenticate] Password:
david@Despliegue:~$ sudo chmod -R 755 /var/www/running-zones
david@Despliegue:~$ ls -la /var/www/running-zones/
total 976
drwxr-xr-x 4 www-data www-data   4096 Nov  5 19:35  .
drwxr-xr-x 6 root     root       4096 Nov  5 18:26  ..
-rwxr-xr-x 1 www-data www-data   4186 Nov  5 19:14  02calculadoraBootstrap.php
-rwxr-xr-x 1 www-data www-data   1147 Nov  5 19:35  04ArrayBootstrap.php
-rwxr-xr-x 1 www-data www-data 491279 Nov  5 18:20 'Instalaci'$'\303\263''n y configuraci'$'\303\263''n
 NGINX.odt'
-rwxr-xr-x 1 www-data www-data 311224 Nov  5 18:20 'Instalaci'$'\303\263''n y configuraci'$'\303\263''n
 NGINX.pdf'
-rwxr-xr-x 1 www-data www-data 165205 Nov  5 18:20 'Pr'$'\303\241''ctica 2.1 Despliegue de NGINX  PHP F
drwxr-xr-x 4 www-data www-data   4096 Nov  5 18:49  mi-app-php
drwxr-xr-x 2 www-data www-data   4096 Nov  5 18:37  running_zone
david@Despliegue:~$
```

Comprobar conexion

```
david@Despliegue:~$ ls -l /run/php/
total 4
lrwxrwxrwx 1 root     root     30 Nov  5 17:41 php-fpm.sock -> /etc/alternatives/php-fpm.sock
-rw-r--r-- 1 root     root      5 Nov  5 17:41 php8.5-fpm.pid
srw-rw---- 1 www-data www-data  0 Nov  5 17:41 php8.5-fpm.sock
```

Hacer Copia de seguridad

```
david@Despliegue:~$ sudo cp /etc/nginx/sites-enabled/nginx-demo /etc/nginx/nginx-demo.backup.conf
```

Editar configuracion

```
 GNU nano 8.7.1                                  /etc/nginx/sites-enabled/nginx-demo
server {
    listen 8080 default_server;
    listen [::]:8080 default_server;

    server_name _;

    root /var/www/nginx-demo/html;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }

    location /running-zones/ {
        root /var/www;
        index 02formCalc.php;
        try_files $uri $uri/ =404;
    }

    location ~ ^/running-zones/.+\.php$ {
        root /var/www;
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/run/php/php8.5-fpm.sock;
    }
}
```

Comprobar que la configuración es correcta

```
david@Despliegue:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
david@Despliegue:~$
```

Comprobar el firewall

```
david@Despliegue:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
80/tcp                     ALLOW       Anywhere
22/tcp                     ALLOW       Anywhere
8080/tcp                   ALLOW       Anywhere
80/tcp (v6)                ALLOW       Anywhere (v6)
22/tcp (v6)                ALLOW       Anywhere (v6)
8080/tcp (v6)              ALLOW       Anywhere (v6)
```

Reiniciar los servicios

```
david@Despliegue:~$ sudo systemctl restart nginx
david@Despliegue:~$ sudo systemctl restart php8.5-fpm
```

```
david@Despliegue:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-11-12 17:10:44 UTC; 8s ago
 Invocation: 3924888c1d5243d3966b2448b09c8ae6
       Docs: man:nginx(8)
    Process: 2082 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, sta
    Process: 2083 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUC
   Main PID: 2085 (nginx)
      Tasks: 4 (limit: 7397)
     Memory: 3.8M (peak: 4.3M)
        CPU: 22ms
     CGroup: /system.slice/nginx.service
             ├─2085 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─2086 "nginx: worker process"
             ├─2087 "nginx: worker process"
             └─2088 "nginx: worker process"
```

```
david@Despliegue:~$ sudo systemctl status php8.5-fpm
● php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager
     Loaded: loaded (/usr/lib/systemd/system/php8.5-fpm.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-11-12 17:10:48 UTC; 1min 20s ago
 Invocation: a1a1120f6dc1421591e82b0e33986fca
       Docs: man:php-fpm8.5(8)
    Process: 2105 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/p
   Main PID: 2102 (php-fpm8.5)
     Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0.00req/sec"
      Tasks: 3 (limit: 7397)
     Memory: 9.8M (peak: 11.8M)
        CPU: 42ms
     CGroup: /system.slice/php8.5-fpm.service
             ├─2102 "php-fpm: master process (/etc/php/8.5/fpm/php-fpm.conf)"
             ├─2103 "php-fpm: pool www"
             └─2104 "php-fpm: pool www"

Nov 12 17:10:48 Despliegue systemd[1]: Starting php8.5-fpm.service - The PHP 8.5 FastCGI Process Manage
Nov 12 17:10:48 Despliegue systemd[1]: Started php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager
lines 1-18/18 (END)
```

Comprobar NGINX con curl

```
david@Despliegue:~$ curl -I http://localhost:8080
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Thu, 12 Nov 2026 17:12:39 GMT
Content-Type: text/html
Content-Length: 230
Last-Modified: Thu, 05 Nov 2026 16:13:40 GMT
Connection: keep-alive
ETag: "6aecab34-e6"
Accept-Ranges: bytes

david@Despliegue:~$
```

Esto comprueba si NGINX responde

Después probar directamente la aplicación

```
david@Despliegue:~$ curl -I http://localhost:8080/running-zones/04ArrayBootstrap.php
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Thu, 12 Nov 2026 17:13:01 GMT
Content-Type: text/html; charset=UTF-8
Connection: keep-alive

david@Despliegue:~$
```

Ahora comprobar que PHP se interpreta

```
david@Despliegue:~$ curl -s http://localhost:8080/running-zones/04ArrayBootstrap.php | head -20
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Ejercicio 4</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="styleshee
384-sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB" crossorigin="anonymous">

</head>
<body class="bg-secondary">

    <h3>Mostrar el primer día </h3>Lunes<h3>Mostrar consecutiva </h3>Lunes, Martes, Miércoles, Jueves,
Domingo,     <h3>Mostrar consecutiva en forma de lista</h3>
    <ol class="list-group list-group-numbered">
    <li class='list-group-item bg-primary text-white'>Lunes</li><li class='list-group-item bg-primary t
</li><li class='list-group-item bg-primary text-white'>Miércoles</li><li class='list-group-item bg-prir
ueves</li><li class='list-group-item bg-primary text-white'>Viernes</li><li class='list-group-item bg-p
'>Sábado</li><li class='list-group-item bg-primary text-white'>Domingo</li></ol><br>    </ol>


</body>
</html>david@Despliegue:~$
```

Comprobación del Despliegue

